When the model is wrong

Optimization · Level 4 · Lesson 3



Today we will

- Find the rules a model is missing by testing its answer with people.
- Check whether a plan jumps when a shaky number changes.
- Write a model’s assumptions down so others can check them.

Remember

- A model obeys only its rules: wrong answers usually mean missing rules.
- Change the numbers you’re least sure of; a jumpy plan needs a second look.
- OPTIMAL means best for the model, not ready for the world.

Break the model

1. Warm-up (5 min): Would you eat this lunch every day?
2. Teach (7 min): A model obeys only its rules.
3. Break it (12 min): Groups list every objection.
4. Fix it (10 min): Turn objections into rules.
5. Assumptions (6 min): Write what the model ignores.

No lunch at school? Use the timetable instead: show a timetable with no clashes but maths always in the last period, and ask pupils to break it.

Let’s discuss

1. Who in our school would notice a missing rule first, and why?
2. Which numbers in our plans are we least sure of?
3. Why might a perfect solver still give a bad plan?

Exit ticket

- Write one missing rule for the cheapest lunch, as a sentence a model could check.